Some Nodes might not be required to implement communication using group keys, in which case they MAY omit the *Global Group Encrypted Message Counters*. In contrast to the *Global Unencrypted Message Counter*, Nodes are required to persist the *Global Group Encrypted Message Counters* in durable storage. In particular, Nodes are required to ensure that the value of the *Global Group Encrypted Message Counters* never rolls back and that it is monotonic within the bounds of its range for its use for a given group key. A Node SHALL randomize the initial value of this counter on factory reset per Section 4.6.1.1, “Message Counter Initialization”.

While *Global Group Encrypted Message Counters* are shared by many group keys to generate nonces, rollover is not an issue as long as the Epoch Key that generates each operational group key rotates frequently enough.

NOTE	If a nonce is duplicated for a given key, the security consequences are isolated only to the specific key with which the duplicate nonce occurred — a key that has not been updated prior to rollover and has been presumably abandoned or aged out.

4.6.2. Secure Session Message Counters

A *Secure Session Message Counter* is a per-session, 32-bit, ephemeral counter that is used by the encoding of any encrypted messages using an associated session key. Each peer in a Secure Unicast Session SHALL maintain its own message counters, with independent counters being used for each unique session used. *Session Message Counters* SHALL exist for as long as the associated security session is in effect. A Node SHALL randomize the initial value of this counter on session establishment per Section 4.6.1.1, “Message Counter Initialization”.

The *Secure Session Message Counter* history window SHALL be maintained for the lifetime of the session, and SHALL be deleted at the same time as the session keys, when the session ends.

Sessions SHALL be discarded and re-established before any *Secure Session Message Counter* overflow or repetition occurs.

4.6.3. Check-In Counter

The Check-In Counter is an unsigned 32-bit counter used during the encryption process of the Check-In Protocol. The only purpose of the Check-In Counter is during the encryption process of the Check-In Protocol.

Each Check-In Protocol use-case implemented by a device SHALL be associated with a specific Check-In counter to be used for each node identity (pair of fabric and node identifier). This MAY be a single counter across all node identities, or MAY be one counter per node identity, or anything in between, as long as each node identity uses a single specific instance of the counter. The device SHALL randomize the initial value of the counter on factory reset per Section 4.6.1.1, “Message Counter Initialization”.

The Check-In Counter SHALL be monotonically increased each time a Check-In message is sent. This monotonicity guarantee SHALL be preserved across idle and active states. The Check-In Counter can roll over to zero when it exceeds the maximum value of the counter (32-bits). Nodes are required to persist the *Check-In Counter* in durable storage.